UDDIPA DATTUDU CHAVALI

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ACADEMIC & RESEARCH PROFILE

Electronics and Communication Engineering undergraduate focused on VLSI, RF CMOS IC design, and analog circuit design, with hands-on research experience in RFIC and CMOS image-sensor circuits. Experienced in transistor-level circuit design, simulation, and analysis using Cadence Virtuoso, SPICE, and Verilog, with additional experience in Python-based EDA automation. Interested in developing practical, high-performance integrated-circuit solutions across analog, RF, and mixed-signal domains.

ACADEMIC BACKGROUND

Madanapalle Institute of Technology and Science — Andhra Pradesh, India
Bachelor of Technology in Electronics and Communication Engineering; CGPA: 9.37 — *July 2025 – Expected May 2028*

ESC Government Polytechnic College — Andhra Pradesh, India
Diploma in Electronics and Communication Engineering; Percentage: 95.15% — *Aug 2022 – May 2025*

TECHNICAL SKILLS

Core Competencies: Analog Circuit Design, Radio Frequency CMOS Integrated Circuit Design, DC/AC/Transient Simulations.
EDA Tools: Cadence Virtuoso, Altium Designer (Basic PCB), NGSPICE, NI Multisim, MATLAB.
Programming & HDLs: Verilog HDL, Python (Currently learning Data Science & Basic Automation), C language.

ACADEMIC, RESEARCH AND INDUSTRIAL EXPERIENCE

National Institute of Technology (NIT) Calicut — Remote
Intern - Analog IC Design for Image Sensors — *July 2026 – Present*

- Collaborating on remote research under the supervision of Dr. Bhuvan B, focusing on the study and design of Programmable Gain Amplifiers (PGAs) and Single-Slope ADCs for CMOS camera image sensors.
- Working on the transistor-level modeling and simulation of the converter blocks to optimize gain, noise, and power for high-resolution imaging applications.

National Institute of Technology (NIT) Calicut — Kerala, India
Summer Research Intern - RFIC Design for Anti-Drone Radar — *May 2026 – July 2026*

- Designed and simulated an unconditionally stable 9.0 GHz cascode Low Noise Amplifier (LNA) in a 180nm CMOS process using Cadence Virtuoso under the guidance of Dr. Karthik R, achieving a forward voltage gain (S_{21}) of 11.06 dB and a Noise Figure (NF) of 1.41 dB.
- Engineered an active LC L-match output network and multi-finger gate topologies (30 fingers of 5.0 μm) to neutralize the Miller effect, ensuring precision 50 Ω impedance matching ($S_{11} = -17.06$ dB, $S_{22} = -23.27$ dB).
- Verified high power efficiency via ADE-L DC operating point extraction, yielding a low power consumption (P_{DC}) of 14.13 mW and an exceptional Figure of Merit (FoM) of 21.05 GHz/mW for X-band receiver front-ends.
- Actively collaborated with the research group, presenting weekly simulation data, design trade-offs, and performance metrics to the principal investigator to iteratively optimize the RF front-end architecture.

Southern Power Distribution Company of A.P. Limited (APSPDCL) — Andhra Pradesh, India
Network Intern in TELECOM Department (Industrial Training as part of my Diploma) — *May 2024 – Nov 2024*

- Completed a 6-month industrial training focused on telecommunications and the physical layer of networking.
- Configured and troubleshot enterprise hardware, including Cisco routers, SD-WAN setups, and 24-port HP/D-Link switches to keep power grid data flowing.
- Gained practical experience with physical infrastructure by crimping Cat6 cables, punching down keystone jacks, and observing the troubleshooting of fiber optic (OFC) links.

ENGINEERING PROJECTS

MINI-ECU: Bare-Metal Embedded Drive Control System | *STM32F401, Bare-Metal C* — Oct 2026

- Contributed to selected bare-metal C code blocks for potentiometer-based motor speed control, using the STM32F401 ADC input and mapping the converted value to PWM duty cycle.
- Studied the associated ADC and timer/PWM peripheral registers and the signal path from potentiometer position to motor-speed command, building practical experience with register-level embedded programming.
- Developed an understanding of the overall MINI-ECU architecture, including finite-state-machine control, UART movement commands, status LEDs, emergency-stop handling, and reset/re-enable flow.

Dual-Input LiPo Battery Charger PCB | *Altium Designer* Aug 2026

- Designed a custom 2-layer PCB for a lithium-ion battery charger, including dual power inputs (Micro-USB and DC barrel jack) safely separated by input OR-ing diodes.
- Used a bottom copper pour to connect all grounds and keep the board cool.
- Added a physical solder jumper to let users switch between 100mA and 500mA charging speeds, mapped 3D component bodies to check clearances, and printed clear silkscreen safety warnings.

Automated CMOS Inverter Characterization | *Python, NGSPICE, Pandas* July 2026 – Aug 2026

- Wrote a Python automation script that runs NGSPICE silently in the background, automatically sweeping 180nm CMOS transistor widths from 1.0 μm to 5.0 μm. Used Pandas to extract and parse the raw simulation data directly from text files, replacing manual GUI testing to efficiently analyze Voltage Transfer Characteristics (VTC).
- Graphed the shifting switching thresholds using Matplotlib to visually prove how changing the PMOS size alters the strength of the pull-up network.

180nm 6T SRAM Cell Design and Characterization | *Analog VLSI Design* July 2026

- Designed a 6T SRAM cell in a 180nm CMOS process using Cadence Virtuoso, optimizing transistor sizing (Cell Ratio = 2.0, Pull-Up Ratio = 0.7) for balanced read/write stability.
- Extracted Voltage Transfer Characteristics (VTC) and applied Seevinck's method via a custom MATLAB script to calculate a robust Read Static Noise Margin (SNM) of 309.0 mV.
- Performed transient and DC analyses to validate energy efficiency, achieving an active dynamic power of 86.39 μW (at a 40ns period) and an ultra-low static leakage of 24.1 pW.

HONORS & AWARDS

1st Prize, Embedded Systems Design track – Maven Silicon Silicon Sprint 2026 | *STM32, Bare Metal Embedded C* Oct 2026

- Won first prize as part of Team COREX in the Embedded Systems category at the Maven Silicon Silicon Sprint semiconductor hackathon.
- Collaborated in a four-member team to develop a bare-metal MINI-ECU drive-control solution on the STM32F401 and demonstrate its embedded control workflow.

1st Prize (INR 25,000), Innovation Nexus 2026 Hackathon | *MITS, PathVision & IITTNiF* Aug 2026

- Awarded 1st place out of 50 competing teams during a rigorous 16-hour hackathon for solving a targeted industry problem statement to develop a "Real-Time Agricultural Transport Condition Monitoring System".
- Led project research, feasibility analysis, and architecture study, ultimately pitching the solution to the jury.
- Programmed an ESP32 microcontroller and physically integrated multi-modal sensors (DHT11, MQ-4, GPS) to build the IoT data ingestion pipeline.

National Level Technical Symposium Wins | *Multiple Institutions* 2024 – 2026

- **2nd Position, "Circuit Detective":** ECLECTICA 2K26 at MITS, Madanapalle (April 2026). Awarded a cash prize.
- **Double 1st Prize, "Circuit Hunt" & "Technical Quiz":** E-Merge 2K24 at JNTUACEA (Nov 2024). Awarded a total of INR 2000 cash prize for winning both events.
- **1st Prize, Paper Presentation:** Ripple National Symposium at RGM College of Engineering, Nandyal. Awarded an INR 1000 cash prize.

Consistent Academic Excellence & Campus Awards | *ESC Govt Polytechnic College* 2023 – 2025

- Consistently ranked at the top of the cohort across all semesters (2023–2025), receiving multiple faculty-awarded merits and cash prizes for highest examination results.
- Secured 1st and 2nd place finishes in consecutive campus-wide technical quizzes (2023).

RELEVANT COURSEWORK

Analog Circuits; Digital System Design (Verilog); Electronic Devices & Circuits; Network Theory; Python Programming; PCB Design; Introduction to Quantum Technologies and Applications.

CERTIFICATIONS

Microsensors and Nanosensors (In Progress) | *NPTEL / IIT Guwahati* July 2026 – Oct 2026

System Design Through Verilog | *NPTEL / IIT Guwahati* July 2026 – Sept 2026

- Studied advanced digital design concepts including behavioral modeling, data flow modeling, and the synthesis of combinational and sequential logic.

VLSI for Beginners | *NIELIT Calicut (MeitY C2S Project)* Aug 2026

Introduction on Intellectual Property to Engineers | *NPTEL / IIT Kharagpur* Jan 2026 – Mar 2026

- Gained essential knowledge of patent filing procedures, trade secrets, and IC Layout Design protection frameworks.

Understanding Incubation and Entrepreneurship | *NPTEL / IIT Bombay* Jan 2026 – Apr 2026

- Studied technology-led entrepreneurship, customer discovery processes, and the Lean Canvas Model to understand the framework for converting technical solutions into viable businesses.

WORKSHOPS & INDEPENDENT STUDY

Radio-Frequency Integrated Circuits (Independent Audit) | *NPTEL / IIT Kanpur* July 2026 – Present

- Auditing coursework focused on wireless receiver/transmitter architectures, studying RF metrics, LNA design, mixers, and Power Amplifier (PA) architectures.

MAC 2026 – Maven Alumni Conference | *Maven Silicon* Oct 2026

- Attended industry sessions and a panel discussion with six professionals from different companies, gaining perspectives on the VLSI and semiconductor ecosystem, Agentic AI, AI orchestration, and skills expected of future engineers.
- Reflected on the value of strong engineering fundamentals, hands-on implementation, problem-solving, clear communication, and continuous learning as technology and engineering workflows evolve.

Seminar: VLSI Testing and Physical Design | *MITS ISTE Student Chapter* Feb 2026

Seminar: VLSI and Electromagnetic Industry Practices | *MITS IETE Students Forum* Feb 2026

Guest Talk: India's Semiconductor Revolution & VLSI | *MITS Alumni Association* Aug 2025